

Temporal LP economics — report

Model: v5 options-seller (auditor-corrected, faithful). Brainstorm / non-core. Magnitudes illustrative.

Thesis: the LP **sells the perpetual-option book** (bands), delta-hedged on Hyperliquid → **short-vol**. It earns the funding-carry priced at the **calibration** vol and pays the **realized**-vol gamma bleed ($G = (m\gamma) (m\gamma + 1) / 2$). **It profits when realized vol < calibration vol** — not "fees – LVR".

Model view — BASE net APR (rows = realized vol, cols = calibration vol · green = profit / red = loss)

RV \ σ_{cal}	30%	50%	70%	90%	110%
20%	16%	64%	136%	232%	352%
40%	-20%	28%	100%	196%	316%
60%	-80%	-32%	40%	136%	256%
80%	-164%	-116%	-44%	52%	172%
100%	-272%	-224%	-152%	-56%	64%

Green upper-right = calmer than priced. Red lower-left = spike / undercalibration. m amplifies both.

Assumptions (entry-496 calibrated; yellow=editable)

gamma / m / LP_leverage	2.0 / 1.0x / 1.0x
daily_turnover	0.30
fee_bps	2.5
hedge_exec_cost	1.0%/yr
order_mgmt_cost	0.5%/yr
HLP_base_APR	10%
$G = (m\gamma)(m\gamma + 1)/2$	3.0

Scenario: RV 60%, σ_{cal} 70%

Carry income (+, at σ_{cal})	147.0%
Vol cost / gamma bleed (LVR-equiv, RISK 1)	-108.0%
= options_net	39.0%
Fee income	2.7%
Hedge+exec+order-mgmt	-1.5%
BASE net	40.2%
HLP layer	+10.0%
RESTAKED net	50.2%

Point-by-point (entry 496)

1 imperfect hedge + execution	modeled: 1.0%/yr
2 turnover 0.2–0.4×	0.30 → fee income ~2.7%/yr (minor)
3 2.5 bps	applied
4 vol cost (LVR-equiv) + poor calibration	split & shown; well-calibrated ≈ +1%, spike/miscalibrate → deep red
5 limit-order placement/updating	modeled: 0.5%/yr
6 principal protection (treasury LPs)	SHELVED — no action

Bottom line

Short-vol business: green in calm/well-priced regimes, red in spikes. With realistic fees (~2.7%) & costs (~1.5%) and **no calibration edge, net ≈ +1%/yr** — the return **requires calibrating vol above realized**; the risk is miscalibration or a spike. Open: real funding law (update-2) sets the carry; coefficient needs calibration.